

CILFYNYDD, United Kingdom

WMO: 036140

Lat: 51.633N

Lon: 3.300W

Elev: 636

StdP: 14.36

Time Zone: 0.00 (EUW)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.5	27.3	20.9	16.0	27.5	23.5	18.2	30.0	33.2	46.6	29.5	45.2	4.8	70	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	78.0	64.3	74.2	62.2	70.8	60.9	65.9	74.9	63.8	71.0	62.1	68.1	6.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	87.5	68.3	61.2	82.6	65.8	59.7	78.2	64.1	30.9	75.2	29.3	71.5	28.1	68.3	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.7	22.2	19.4	DB	20.7	82.3	4.1	4.2	17.7	85.3	15.3	87.8	13.1	90.1	10.1	93.2
			WB	19.6	69.0	4.1	2.0	16.7	70.4	14.2	71.6	11.9	72.7	8.9	74.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	39.9	38.9	42.9	45.5	52.7	56.8	60.7	59.4	54.6	49.8	44.2	41.2	(6)
	DBStd	9.08	5.69	6.12	4.37	4.97	5.50	5.57	4.53	4.78	3.92	4.95	5.49	5.76	(7)
	HDD50	1568	312	312	222	154	30	5	0	0	6	66	184	277	(8)
	HDD65	5899	777	731	685	585	382	256	149	187	312	471	625	739	(9)
	CDD50	1185	0	0	2	19	114	208	332	293	145	60	9	3	(10)
	CDD65	40	0	0	0	0	1	8	16	14	1	0	0	0	(11)
	CDH74	330	0	0	0	0	13	68	129	120	0	0	0	0	(12)
	CDH80	46	0	0	0	0	0	7	13	26	0	0	0	0	(13)
(14) Wind	WSAvg	7.1	9.0	8.7	9.1	7.5	6.4	5.8	5.8	5.4	5.1	6.7	6.5	8.9	(14)
(15) Precipitation	PrecAvg	51.3	6.1	4.3	3.3	3.3	3.4	3.0	3.1	3.8	3.8	5.5	5.7	6.1	(15)
	PrecMax	62.4	10.5	10.4	6.9	8.0	6.7	7.6	8.6	9.1	7.8	10.2	11.0	11.5	(16)
	PrecMin	34.7	0.4	0.4	0.8	0.5	0.4	0.6	0.7	0.6	0.7	1.6	2.7	1.6	(17)
	PrecStd	7.2	2.6	2.8	1.7	1.9	1.5	1.7	1.9	2.0	1.8	2.4	2.2	3.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	52.5	58.5	68.5	75.9	80.4	81.8	83.5	72.0	63.7	56.0	53.9	(19)
		MCWB	49.8	49.7	51.1	57.6	61.3	64.7	65.3	67.6	61.3	57.1	52.5	52.0	(20)
	2%	DB	50.3	50.5	53.3	62.4	71.5	75.6	77.7	77.4	66.8	60.2	54.8	52.9	(21)
		MCWB	48.7	48.2	48.1	53.0	59.5	62.2	64.3	65.3	59.2	55.9	53.1	51.2	(22)
	5%	DB	49.1	48.8	50.4	56.8	67.2	71.8	73.8	71.6	64.0	58.4	53.5	51.5	(23)
		MCWB	47.6	46.9	47.3	50.4	57.0	60.7	62.3	62.1	58.2	54.9	51.9	50.1	(24)
	10%	DB	47.8	47.4	49.0	53.6	63.1	67.5	70.0	67.9	61.9	57.0	52.2	50.0	(25)
		MCWB	46.2	45.5	46.5	48.8	54.9	58.6	61.2	60.5	56.9	54.1	50.4	48.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	50.7	52.3	58.3	62.5	66.8	67.4	69.6	62.9	58.9	55.2	52.8	(27)
		MCDB	51.0	51.7	56.3	66.9	72.9	75.7	76.2	78.3	66.9	61.4	55.3	53.2	(28)
	2%	WB	49.2	48.6	49.5	54.0	60.1	63.7	65.6	66.3	61.1	57.2	53.6	51.7	(29)
		MCDB	50.0	49.7	51.6	59.6	70.2	72.6	74.6	75.5	64.9	59.0	54.3	52.4	(30)
	5%	WB	47.8	47.5	48.3	51.7	57.8	61.4	63.7	63.6	59.4	55.8	52.2	50.2	(31)
		MCDB	48.8	48.6	49.8	55.3	66.2	70.0	71.1	69.5	62.9	57.7	53.2	51.1	(32)
	10%	WB	46.4	45.7	47.3	49.7	55.5	59.4	62.0	61.5	57.8	54.6	50.7	48.6	(33)
		MCDB	47.7	47.1	48.7	52.9	61.6	66.6	68.0	66.0	61.1	56.6	52.0	49.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.4	9.7	12.3	13.9	13.6	13.2	13.6	12.5	10.0	8.8	7.3	(35)
		MCDBR	8.8	8.9	11.7	17.9	20.5	20.7	21.2	20.6	16.2	11.2	8.5	6.7	(36)
	5% WB	MCWBR	9.2	7.9	8.3	10.6	10.6	10.8	10.5	11.0	10.4	8.5	7.5	7.1	(37)
		MCWBR	8.5	8.4	9.7	14.8	19.7	18.3	17.4	18.0	13.3	9.5	7.7	6.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.325	0.339	0.370	0.390	0.395	0.386	0.385	0.370	0.362	0.349	0.323	0.314	(40)
		taud	2.344	2.376	2.294	2.260	2.284	2.343	2.350	2.407	2.425	2.429	2.400	2.320	(41)
	Ebn at Noon		214	247	259	268	272	276	274	272	261	240	214	196	(42)
		Edh at Noon	20	26	33	39	40	38	37	33	29	24	19	18	(43)
(44) All-Sky Solar Radiation	RadAvg		246	463	823	1265	1557	1687	1577	1304	983	556	294	191	(44)
	RadStd		22	56	104	145	81	180	153	113	73	49	29	23	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page